Downloaded from geology.gsapubs.org on October 28, 2010

Geology

Paleomagnetic and structural evidence for middle Tertiary counterclockwise block rotation in the Dixie Valley region, west-central Nevada

Mark R. Hudson and John W. Geissman

Geology 1987;15;638-642
doi: 10.1130/0091-7613(1987)15<638:PASEFM>2.0.CO;2

Email alerting services click www.gsapubs.org/cgi/alerts to receive free e-mail alerts when new articles cite this article

Subscribe click www.gsapubs.org/subscriptions/ to subscribe to Geology

Permission request click http://www.geosociety.org/pubs/copyrt.htm#gsa to contact GSA

Copyright not claimed on content prepared wholly by U.S. government employees within scope of their employment. Individual scientists are hereby granted permission, without fees or further requests to GSA, to use a single figure, a single table, and/or a brief paragraph of text in subsequent works and to make unlimited copies of items in GSA's journals for noncommercial use in classrooms to further education and science. This file may not be posted to any Web site, but authors may post the abstracts only of their articles on their own or their organization's Web site providing the posting includes a reference to the article's full citation. GSA provides this and other forums for the presentation of diverse opinions and positions by scientists worldwide, regardless of their race, citizenship, gender, religion, or political viewpoint. Opinions presented in this publication do not reflect official positions of the Society.

Notes

Geological Society of America

THE GEOLOGICAL SOCIETY OF AMERICA

Downloaded from geology.gsapubs.org on October 28, 2010

Paleomagnetic and structural evidence for middle Tertiary counterclockwise block rotation in the Dixie Valley region, west-central Nevada

Mark R. Hudson
Department of Geology, Colorado School of Mines, Golden, Colorado 80401
John W. Geissman
Department of Geology, University of New Mexico, Albuquerque, New Mexico 87131

ABSTRACT

Paleomagnetic data from late Oligocene to early Miocene ash-flow tuffs at four localities in the northern Dixie Valley region, west-central Nevada, indicate that parts of the crust have rotated counterclockwise by at least 25° and perhaps significantly more in late Cenozoic time. Field relations in White Rock Canyon, Stillwater Range, suggest that rotation (1) was accommodated by right-lateral slip on northwest-trending faults, (2) spanned ash-flow tuff emplacement, and (3) probably ceased before eruption of overlying middle Miocene basalts. Accurate estimates of Cenozoic extension, as well as evaluation of earlier Mesozoic structures, must include the strain partitioned into rotation in the area.

INTRODUCTION

Recent research on the Cenozoic tectonic development of the Basin and Range province, western North America, has focused on highly extended rocks tilted to steep dips above gently dipping detachment faults (e.g., Proffett, 1977; papers in Crittenden et al., 1980; Wernicke, 1981; Geissman et al., 1982; Miller et al., 1983). Regions where Tertiary and conformable underlying strata dip less steeply are thought to be less extended than regions that have high stratal dips. Significant crustal extension may also be accommodated, however, by strike-slip faulting and attending block rotation (Freund, 1974; Ron et al., 1984), resulting in little stratal dip. Paleomagnetic and fault-slip data from the northern Dixie Valley region suggest that middle Tertiary extension was largely accommodated by body rotation of crustal blocks about vertical axes and attending strike-slip motion on northwest-trending faults, rather than solely by normal faulting. This deformation predated widespread basaltic volcanism in middle Miocene time. Recognition of it has implications for evaluation of modes of Cenozoic extension as well as Mesozoic and late Paleozoic deformation in this part of the Basin and Range province.

GEOLOGIC SETTING

The localities studied are in the Stillwater and Clan Alpine ranges bounding Dixie Valley (Fig. 1). The oldest rocks exposed (Fig. 1) are lower Mesozoic metasedimentary strata, formed in a Mesozoic marine province, and are intruded by the Middle Jurassic Humboldt lopolith gabbroic complex (Oldow, 1984; Page, 1965; Speed, 1976). Mesozoic rocks are overlain by Tertiary volcanic and sedimentary strata. Locally preserved early Oligocene andesitic lavas that have K-Ar ages of 35 Ma (Riehle et al., 1972) are overlain by a regionally extensive sequence of Oligocene to early Miocene ash-flow tuffs. K-Ar ages of equivalent tuffs exposed in the southern Clan Alpine Range and other surrounding ranges span 32 to 22 Ma (Riehle et al., 1972; Speed, 1976; Burke and McKee, 1979); in the northern Dixie Valley area the tuffs have not been dated. In the Stillwater Range, detrital sedimentary rocks overlie the ash-flow tuffs and are capped by Miocene basalts, which have yielded K-Ar ages of 13 to 17 Ma (Nosker, 1981).

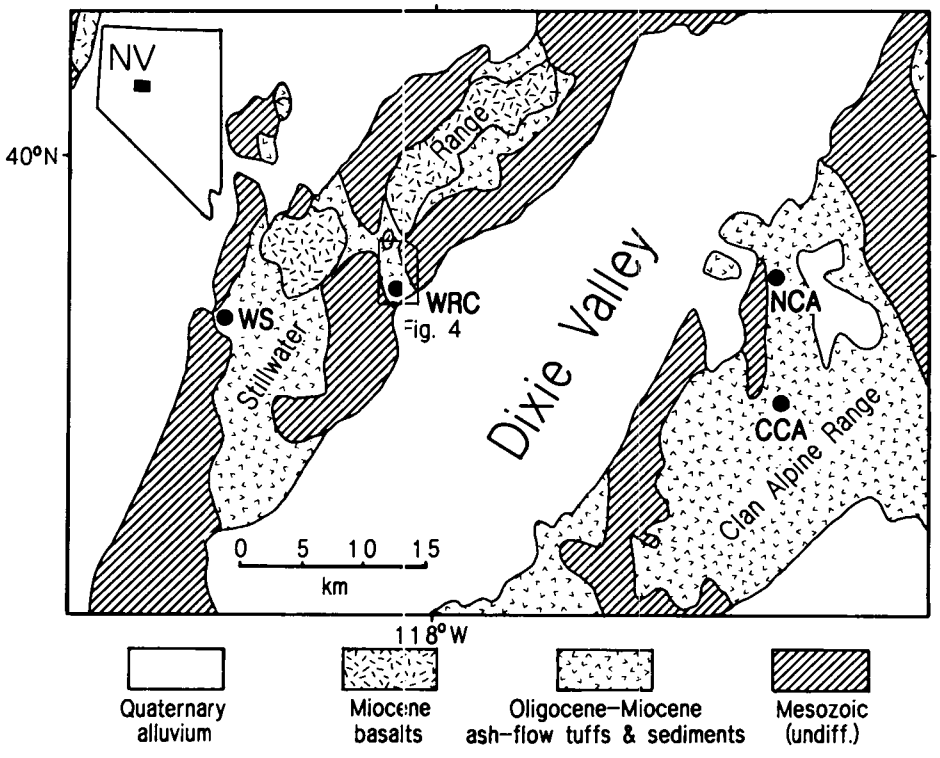


Figure 1. Localities studied in Stillwater and Clan Alpine ranges, northern Dixie Valley region, west-central Nevada. NCA = northern Clan Alpine Range; CCA = central Clan Alpine Range; WRC = White Rock Canyon area, Stillwater Range; WS = western Stillwater Range.

Downloaded from geology.gsapubs.org on October 28, 2010

Layered Mesozoic strata were thrust and folded during the Jurassic–Early Cretaceous emplacement of the Luning-Fencemaker allochthon (Speed, 1976; Oldow, 1984). Late Cretaceous plutons crosscut all Mesozoic strata. Late Miocene, high-angle normal faults produced the horst-graben structure controlling present physiography (Okaya and Thompson, 1985). Analysis of the style of middle Tertiary deformation observed in Oligocene and early Miocene ash-flow tuffs is the subject of this paper.

PALEOMAGNETISM

Discordant declinations of paleomagnetic vectors from the Jurassic Humboldt lopolith (Hudson and Geissman, 1984), implying that the complex had been significantly rotated (–37° ±14° to ±22°, depending on the age of its magnetization) in a counterclockwise manner, prompted evaluation of possible, yet unexpected, components of Tertiary rotation in the Dixie Valley region. Figure 1 shows the localities where samples for paleomagnetic analysis were collected from Tertiary ash-flow tuff cooling units: northern Clan Alpine (NCA; 6 sites, 5 units); central Clan Alpine (CCA; 17 sites, 12 units); White Rock Canyon area (WRC; 18 sites, 13 units); and western Stillwater (WS; 16 sites, 10 units). At the WRC (4 sites, 2 units) and WS (2 sites, 2 units) localities, we sampled diabase sills and a basalt flow whose crosscutting relations indicate age equivalence to the upper ash-flow tuff units. All samples were treated by using either progressive alternating field or thermal demagnetization. Well-grouped magnetizations, which we believe to be thermoremanent magnetizations, were isolated in a straightforward fashion (Fig. 2); alternating field demagnetization was preferred because it was effective in removing occasional secondary magnetizations acquired during lightning strikes. Data from individual sites (Fig. 3) are well grouped (Fisher, 1953, precision parameter [k] values typically exceed 50; a compilation of all site data is available from Hudson or Geissman upon request). In many cases duplicate site directions in cooling units are statistically distinct (at confidence levels greater than 95%, applying the technique of McFadden and Lowes, 1981), either because of errors in assessing differential tilt corrections between sites or because of intra-unit paleomagnetic dispersion that may arise from several sources (e.g., Reynolds, 1977; Geissman et al., 1982). Data from duplicate sites were averaged to obtain unit mean directions (Fig. 3). Magnetizations of individual cooling units are usually statistically distinct at a 95% confidence level from those of adjacent units, suggesting considerable time lapsed between each eruption. Four to eight polarity reversals are recorded in tuff sequences at each locality,

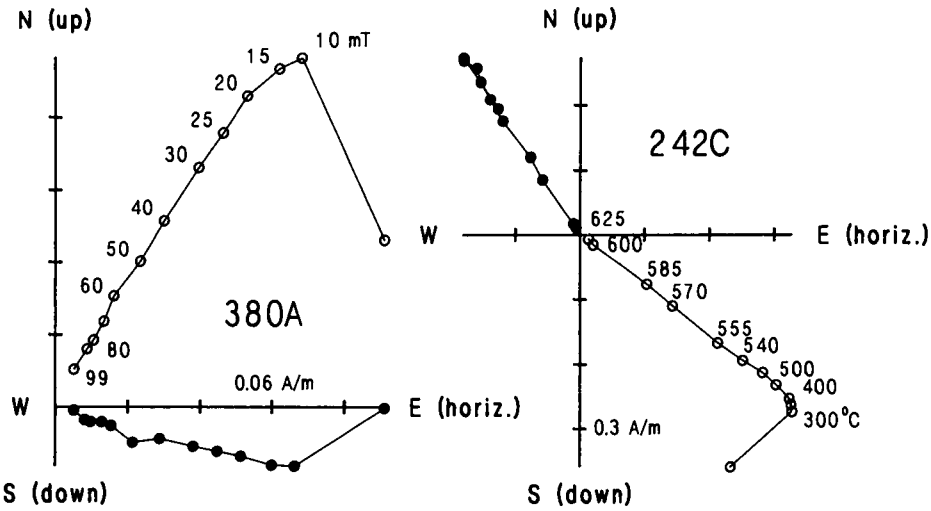


Figure 2. Orthogonal demagnetization diagrams showing behavior of ash-flow tuffs during progressive alternating field and thermal demagnetization. Solid symbols are projections of magnetization vector on horizontal plane; open symbols, vertical plane. Peak demagnetizing fields are given in milliTesla (1 mT = 10 oe) and temperatures in degrees Celsius.

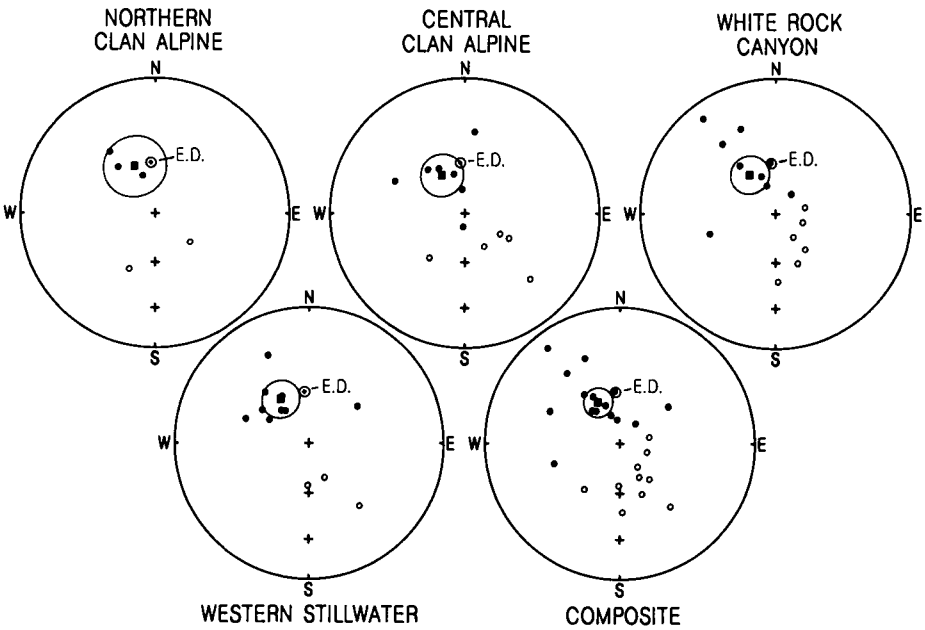


Figure 3. Equal-area projections of unit mean magnetizations from Oligocene–early Miocene ash-flow tuffs at four locations and composite section for Dixie Valley region. Solid symbols are projections on lower hemisphere; open symbols, upper hemisphere. Large square in each projection represents mean paleomagnetic direction; cone of 95% confidence is calculated for each locality. Expected direction (E.D.) and its cone of 95% confidence are also shown.

further confirming that a long geomagnetic time span (as much as several million years) has been sampled.

At each locality the tilt-corrected unit means were averaged to obtain a mean paleomagnetic direction (Table 1). For each locality, the observed mean is discordant (counterclockwise) from an expected middle Tertiary direction (D = 355°, I = +59°, calculated by using the 30 Ma pole of Irving and Irving, 1982). We note that the Irving and Irving (1982) 30 Ma pole is statistically identical, as a confidence level greater than 95%, with the Diehl et al. (1983) 22 to 38 Ma pole. Angular differences in declination (R; Beck, 1980) between observed and expected values are –19° for NCA units, –26° for CCA units, –28° for WRC units, and –28° for WS units. These differences are statistically significant as the 95% confidence level (Beck, 1980; Demarest, 1983) for the CCA, WRC, and WS localities (Table 1). Mean inclinations for the four locations are concordant.

Downloaded from geology.gsapubs.org on October 28, 2010

Not all ash-flow tuff units sampled are present at each sampling locality. To better constrain the middle Tertiary mean paleomagnetic direction, paleomagnetic data from 25 units were assembled from a composite stratigraphic section of tuffs for the region (Fig. 3). Duplicate site means from tuffs sampled at more than one locality were not used in the composite section. For the composite section, the declination discordance (R ±ΔR) is −23° ±15° and inclinations (F ±ΔF = −3° ±7°) are again concordant. An alternative estimate, using the four locality means, results in a mean (Table 1) identical at the 99% confidence level with the composite mean. As expected, the precision estimate is significantly higher for this determination (k value of 688 vs. 12), and the error of the rotation estimate (ΔR) decreases from ±15° to ±7°.

Our data suggest that the ash-flow tuffs and, presumably, underlying basement were rotated counterclockwise about vertical axes. The declination discrepancy observed in data from the Humboldt lopolith (Hudson and Geissman, 1984) thus appears to be at least partly explained by Tertiary crustal rotation. Rotations estimated from the locality or composite means probably underestimate total late Cenozoic rotation because field relations (below) indicate that faulting, and probably rotation, spanned ash-flow tuff emplacement. If the composite section is arbitrarily divided (Table 1), the rotation estimate calculated for the older 13 units (R = −37° ±21°) is notably greater than that from the younger 12 units (R = −11° ±22°), supporting our inference that rotation was concurrent with ash-flow tuff deposition. Discordant paleomagnetic data from the Humboldt lopolith might therefore reflect Tertiary rotation, although the data from the Humboldt Lopolith and extrusive equivalent rocks are still being assessed (Hudson and Geissman, in prep.)

FIELD RELATIONS AND STRUCTURAL ANALYSIS OF THE ASH-FLOW TUFFS

Recognition that parts of the Dixie Valley area were rotated prompted us to search for structures that could have accommodated the observed amount and sense of rotation The geology of the White Rock Canyon area was mapped at a scale of 1:12,000. The area is segmented into structural blocks by major northwest- to north-trending faults and by smaller northeast- to east-trending faults (Fig. 4). Ten mappable ash-flow tuffs were identified in the approximately 1-km-thick section, which occupies a northwest-trending synclinal structure produced by differential tectonic tilting of different structural blocks. Most compaction foliations of tuffs and bedding of intercalated volcanogenic sedimentary strata are tilted about northwest–north-northwest strikes from 20° to the northeast to 60° to the southwest.

Field relations indicate that faulting and tilting spanned deposition of the ash-flow tuff section. Younger tuffs overlap more highly tilted older tuffs with angular unconformity. Some faults cutting older tuffs are overlain by younger tuffs. Miocene sedimentary strata unconformably overlie the ash-flow tuff sequence. Basal parts of the sedimentary strata dip moderately and are slightly offset by several faults offsetting the tuffs, but younger strata overlap these faults and dip very gently. Middle Miocene basalt flows cap the sediments and dip less than 5°. Deformation of the Oligocene–Miocene tuffs waned during sediment deposition and apparently ceased before emplacement of the middle Miocene basalts in the White Rock Canyon area. Within the map area, two north-trending faults that have dip-slip offsets of as much as 1 km juxtapose middle Miocene basalts against lopolith rocks. These faults and those along the range front are the only structures that exhibit large offsets during the youngest phase of high-angle normal faulting.

A deformation model explaining how rotation affected the northern Dixie Valley area should be consistent with observed structures. Although the WRC is a relatively small area, our structural observations are the most detailed in the northern Dixie Valley, and we suggest that general structural styles observed there are characteristic of most of the area, especially where the paleomagnetic data document Tertiary counterclockwise rotation. In the WRC area, both sets of map-scale faults in the tuffs usually exhibit steep (>80°) dips, although some northwest- to north- and northeast-trending faults dip at more moderate angles (40°–65°). Some of the moderately dipping faults of northwest trend cut the tuffs at high angles to bedding, implying that they originally dipped steeply and were later tilted to their present attitude. Faults with northeast to east trends are nowhere offset and typically exhibit less stratigraphic throw than those of the northwest- to north-trending set.

Dip-slip displacements can be estimated from stratigraphic throw across faults. Although the sense and amount of strike-slip displacement are more difficult to constrain and surfaces exhibiting large fault-plane features indicative of direction of offset are rare, we believe that significant right-lateral displacement occurred on many of the northwest-trending faults. On one steeply dipping northwest-trending fault, a 3–4-m-diameter mullion plunges 12° southeast. The stratigraphic throw suggests that the fault underwent right-lateral strike-slip displacement of about 0.75 km. Similar trends and steep dips suggest that several other northwest-trending faults have undergone significant right-slip. Several of the northwest-trending faults curve to the right into more northerly trends. This is consistent with an interpretation that right-lateral slip on northwest-trending segments changes to an increasing component of normal dip-slip displacement on north-trending segments in a westward-directed least principal stress field. Detailed discussion of paleostress determinations is beyond the scope of this paper; it must suffice to say that slip lineation data from 88 small faults in the tuffs have subequal populations of strike-slip, oblique-slip, and dip-slip rakes (Fig. 5). The distribution of small strike-slip faults is of note with respect to possible mechanisms for block rotations implied by the paleomagnetic data. In both in situ and restored orientations, most small strike-slip faults strike northwest and parallel major faults. We show the tilt-corrected distribution in Figure 5 because the northwest maxima become more dominant and the grouping of attitudes of the faults with steep (>80°) dips is improved. Other strike-slip faults strike northeasterly and are parallel to the

Table 1. TILT-CORRECTED PALEOMAGNETIC DATA, DIXIE VALLEY AREA, WEST-CENTRAL NEVADA

Locality	D	I	α_{95}	k	$N/N_o/N_u$	Vector VGP R	ΔR	F	ΔF	AD	AD
White Rock Canyon (WRC)	327	61	11.6	12	22/22/15	−28	20	−2	9	25	35
Western Stillwater Range (WS)	327	58	11.3	11	18/18/12	−28	18	1	9	21	32
Central Clan Alpine Range (CCA)	329	63	12.9	12	17/17/12	−26	23	−4	10	23	38
Northern Clan Alpine Range (NCA)	336	59	18.7	18	6/6/5	−19	30	0	15	19	30
Composite Section - all units	332	62	8.7	12	25 units	−23	15	−3	7	24	35
" " - lower units	318	64	11.3	14	13 units	−37	21	−5	9	22	34
" " - upper units	344	58	13.8	11	12 units	−11	11	1	11	25	34
Locality Means	330	60	2.7	688	4 localities	−25	7	−1	4	4	5

Note: D, I, α_{95}, and k are declination, inclination, cone of 95% confidence, and precision parameter (Fisher, 1953) of mean paleomagnetic direction from groups of tilt-corrected cooling-unit means. $N/N_o/N_u$ = number of sites accepted/sites measured/independent cooling units. Values of rotation (R) and flattening (F) and their 95% confidence levels (ΔR and ΔF, respectively) were calculated using the methods of Beck (1980) and Demarest (1983). Vector AD and VGP AD values are locality/composite direction and virtual geomagnetic pole angular dispersion values, respectively.

Downloaded from geology.gsapubs.org on October 28, 2010

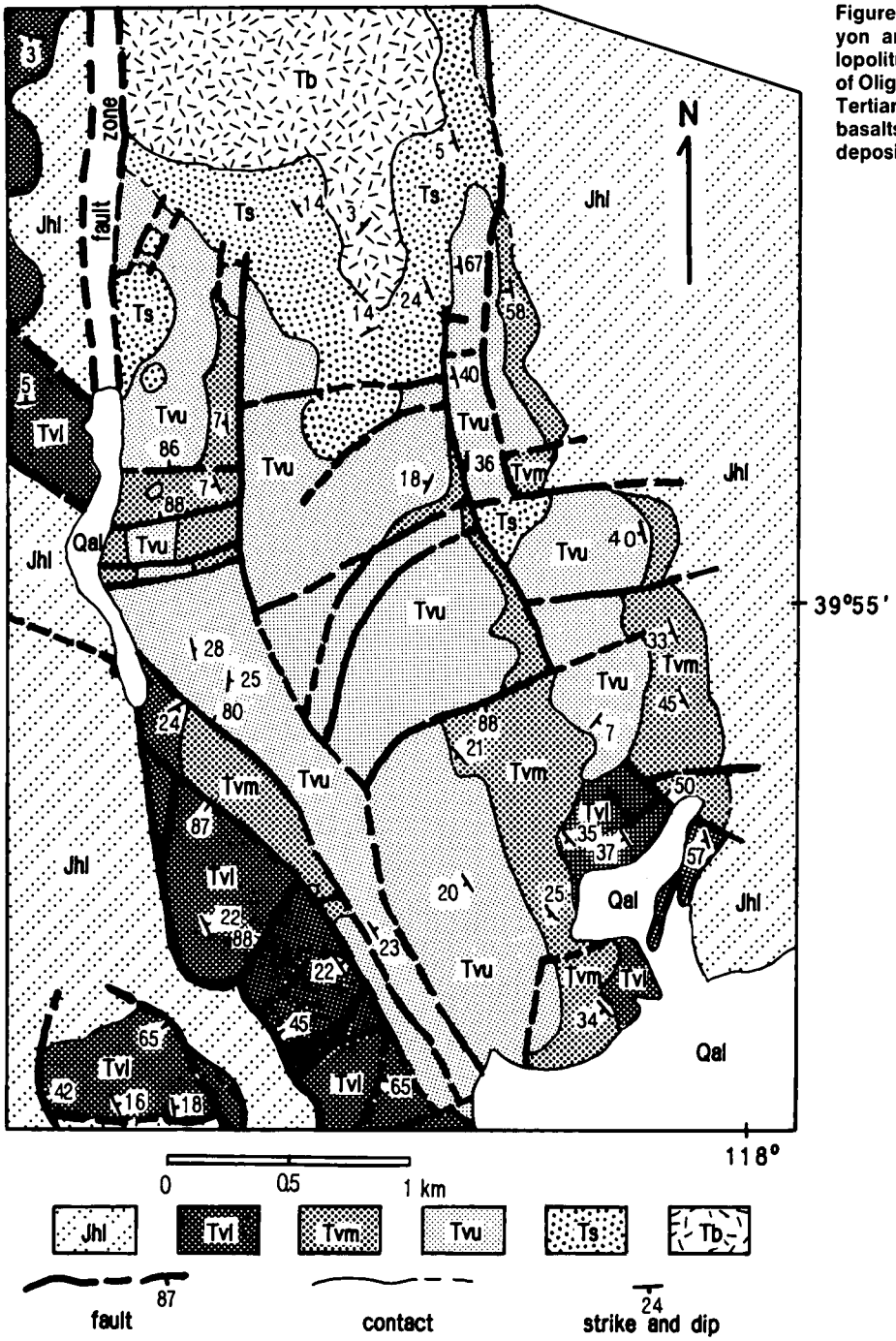


Figure 4. Simplified geologic map of White Rock Canyon area, Stillwater Range. Jhl = Jurassic Humboldt lopolith; Tvl, Tvm, Tvu = lower, middle, and upper parts of Oligocene–early Miocene ash-flow tuff sequence; Ts = Tertiary sedimentary rocks; Tb = middle Miocene basalts; Qal = Quaternary alluvium and landslide deposits.

small, map-scale faults between large, northwest-trending faults. The critical observation is that significant strike-slip motion occurred along map-scale faults; however, many faults with northwest and northeast trends also underwent dip slip and oblique slip (Fig. 5), implying that Cenozoic deformation involved more than strike-slip faulting.

DISCUSSION

The paleomagnetic data strongly suggest that ash-flow tuffs in the Stillwater and Clan Alpine ranges have undergone significant counterclockwise rotation about vertical axes. In models for vertical-axis rotation (e.g., Freund, 1970, 1974; Beck, 1976; Luyendyk et al., 1980; McKenzie and Jackson, 1983, 1986; Ron et al., 1984, 1986; Garfunkel and Ron, 1985), rotation is closely associated with strike-slip faulting. If structures in the WRC area typify those throughout the Dixie Valley region, then we propose that significant right-lateral slip along major northwest-trending faults and block rotation were coeval. Models in which rotation occurs within arrays of steeply inclined, parallel faults (e.g., Freund, 1974; Ron et al., 1984, 1986) address right-lateral slip and attending counterclockwise rotation (Fig. 6). We suggest that slip along northwest-trending right-lateral faults has accommodated counterclockwise rotation. Using the block rotation model of Ron et al. (1984) and assuming a range of allowable boundary orientations (east-west to northwest-southeast) for the rotated region, we find the displacement-to-width ratio on bounding fault

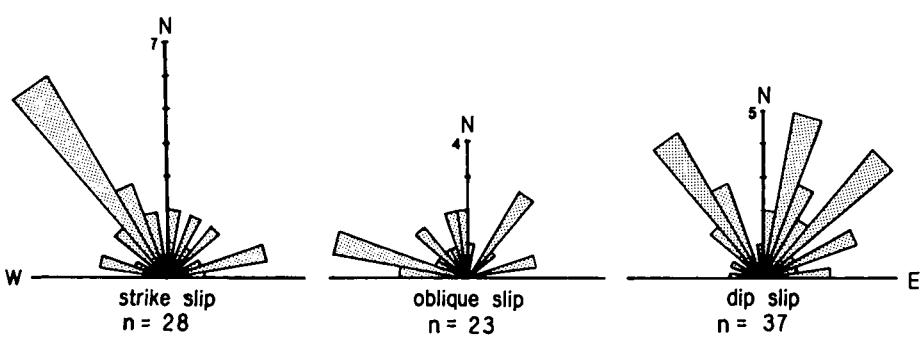


Figure 5. Rose diagram of strikes of small faults with strike-slip (rake <30°), oblique-slip (30° < rake <60°), and dip-slip (rake >60°) motion from White Rock Canyon area, Stillwater Range.

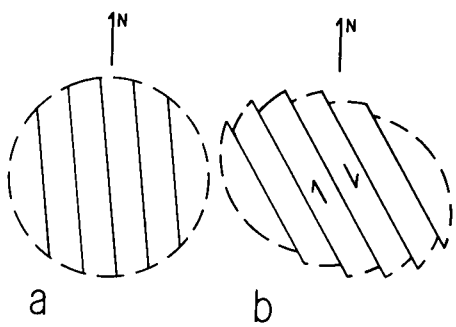


Figure 6. Schematic geometric model to explain observed counterclockwise rotation in northern Dixie Valley region. Model, showing (a) likely initial orientation of strike-slip faults rotated (b) into a more northwesterly trend, does not include components of local tilting.

Downloaded from geology.gsapubs.org on October 28, 2010

blocks ranges from 0.5 to 2, in agreement with field observations, for 25° of counterclockwise rotation. Constraints on the timing of faulting in the White Rock Canyon area imply that rotation probably occurred during emplacement of the tuffs but ceased before eruption of middle Miocene basalts. The apparent decrease in declination discordance, from oldest to youngest tuffs, supports this conclusion. We are currently obtaining data on several middle Miocene basalt sections in the Dixie Valley area to better constrain the age of rotation.

Brittle deformation of the tuffs clearly involved more than rotation of a set of crustal blocks lying between parallel strike-slip faults, as depicted in Figure 6. Tilts of Cenozoic strata about northwest-striking axes within fault blocks and slip data from small faults indicate that dip-slip as well as strike-slip motion occurred along northwest-trending faults. These complexities are expected because orientations of faults bounding rotating blocks constantly change with respect to the stress system under which they formed (Ron et al., 1984; Jackson and McKenzie, 1984; Nur et al., 1986). Furthermore, permutations of the orientations of maximum and intermediate principal stresses, as suggested by Angelier et al. (1985) in the Lake Mead area of southern Nevada, could also have led to intermingled strike-slip and dip-slip motion on faults that have proper orientations. We believe that the overall middle Tertiary deformation in the Dixie Valley region was transtensional, involving phases of strike-slip, oblique-slip, and dip-slip faulting mixed in space and time. Figure 6 is schematic and only strictly applicable to the strain accommodated by the component of strike-slip motion on faults and vertical axis rotation.

CONCLUSIONS

Structural data from the WRC area of the Dixie Valley region imply that 25° or more of counterclockwise rotation of Oligocene and early Miocene ash-flow tuffs, as determined by paleomagnetic data, and underlying rocks accompanied middle Tertiary right-slip along currently northwest-trending faults. The regional extent of the domain of rotation has not been delineated and must be known before accurate estimates of extension partitioned into rotation in this part of the Basin and Range province are afforded. We speculate that rotation has probably been confined to a northwest-trending zone extending across much of Nevada. In the southern Toquima Range, for example, Boden (1986) has described northwest-trending strike-slip faults in ash-flow tuffs that predate latest Cenozoic range-front faults. We are also acquiring paleomagnetic data from Tertiary rocks beyond the Dixie Valley area in hopes of better defining the zone(s) affected by vertical axis rotations.

REFERENCES CITED

Angelier, J., Colletta, B., and Anderson, R.E., 1985, Neogene paleostress changes in the Basin and Range; a case study at Hoover Dam, Nevada-Arizona: Geological Society of America Bulletin, v. 96, p. 347–361.

Beck, M.E., Jr., 1976, Discordant paleomagnetic pole position as evidence of regional shear in the Western Cordillera of North America: American Journal of Science, v. 276, p. 694–712.

——1980, Paleomagnetic record of plate margin tectonic processes along the western edge of North America: Journal of Geophysical Research, v. 85, p. 7115–7131.

Boden, D.R., 1986, Eruptive history and structural development of the Toquima caldera complex, central Nevada: Geological Society of America Bulletin, v. 97, p. 61–74.

Burke, D.B., and McKee, E.H., 1979, Mid-Cenozoic volcano-tectonic troughs in central Nevada: Geological Society of America Bulletin, v. 90, p. 181–184.

Crittenden, M.D., Jr., Coney, P.J., and Davis, G.H., editors, 1980, Cordilleran metamorphic core complexes: Geological Society of America Memoir 153, 490 p.

Demarest, H.H., 1983, Error analysis for the determination of tectonic rotation from paleomagnetic data: Journal of Geophysical Research, v. 88, p. 4321–4328.

Diehl, J.F., Beck, M.E., Jr., Beske-Diehl, S., Jacobson, D., and Hearn, B.C., Jr., 1983, Paleomagnetism of the Late Cretaceous–early Tertiary north-central Montana alkalic province: Journal of Geophysical Research, v. 88, p. 10593–10609.

Fisher, R.A., 1953, Dispersion on a sphere: Royal Society of London Proceedings, v. A217, p. 295–305.

Freund, R., 1970, Rotation of strike-slip faults in Sistan, southeastern Iran: Journal of Geology, v. 78, p. 188–200.

——1974, Kinematics of transform and transcurrent faults: Tectonophysics, v. 21, p. 93–134.

Garfunkel, Z., and Ron, H., 1985, Block rotation and deformation by strike-slip faults, 2. The properties of a type of macroscopic discontinuous deformation: Journal of Geophysical Research, v. 90, p. 8589–8602.

Geissman, J.W., Van der Voo, R., and Howard, K.L., 1982, A paleomagnetic study of the structural deformation in the Yerington district, Nevada, I. Tertiary units and their tectonism: American Journal of Science, v. 282, p. 1042–1079.

Hudson, M.R., and Geissman, J.W., 1984, Preliminary paleomagnetic data from the Jurassic Humboldt lopolith, west-central Nevada: Evidence for thrust belt rotation in the Fencemaker allochthon: Geophysical Research Letters, v. 11, p. 828–831.

Irving, E., and Irving, G.A., 1982, Apparent polar wander paths: Carboniferous through Cenozoic and the assembly of Gondwana: Geophysical Surveys, v. 5, p. 141–188.

Jackson, J., and McKenzie, D., 1984, Active tectonics of the Alpine-Himalayan belt between western Turkey and Pakistan: Royal Astronomical Society Geophysical Journal, v. 77, p. 185–264.

Luyendyk, B.P., Kammerling, M.J., and Terres, R., 1980, Geometric model for Neogene crustal rotations in southern California: Geological Society of America Bulletin, v. 91, p. 211–217.

McFadden, P.L., and Lowes, F.J., 1981, The discrimination of mean directions drawn from Fisher distributions: Royal Astronomical Society Geophysical Journal, v. 67, p. 19–33.

McKenzie, D., and Jackson, J., 1983, The relationship between strain rates, crustal thickening, paleomagnetism, finite strain and fault movements within a deforming zone: Earth and Planetary Science Letters, v. 65, p. 182–202.

——1986, A block model of distributed deformation by faulting: Geological Society of London Journal, v. 143, p. 349–353.

Miller, E.L., Gans, P.B., and Garing, J., 1983, The Snake Range decollement—An exhumed mid-Tertiary ductile-brittle transition: Tectonics, v. 2, p. 239–263.

Nosker, S.A., 1981, Stratigraphy and structure of the Sou Hills, Pershing County, Nevada [M.S. thesis]: Reno, University of Nevada, 60 p.

Nur, A., Ron, H., and Scotti, O., 1986, Fault mechanics and the kinematics of block rotations: Geology, v. 14, p. 746–749.

Okaya, D.A., and Thompson, G.A., 1985, Geometry of Cenozoic extensional faulting: Dixie Valley, Nevada: Tectonics, v. 4, p. 107–125.

Oldow, J.S., 1984, Evolution of a late Mesozoic back-arc fold and thrust belt, northwestern Great Basin, U.S.A.: Tectonophysics, v. 102, p. 245–274.

Page, B.M., 1965, Preliminary geologic map of a part of the Stillwater Range, Churchill County, Nevada: Nevada Bureau of Mines Map 28, scale 1:125,000.

Proffett, J.M., 1977, Cenozoic geology of the Yerington district, Nevada, and implications for the nature and origin of Basin and Range faulting: Geological Society of America Bulletin, v. 88, p. 247–266.

Reynolds, R.L., 1977, Paleomagnetism of welded tuffs of the Yellowstone Group: Journal of Geophysical Research, v. 82, p. 3677–3693.

Riehle, J.R., McKee, E.H., and Speed, R.C., 1972, Tertiary volcanic center, west-central Nevada: Geological Society of America Bulletin, v. 83, p. 1383–1396.

Ron, H., Freund, R., Garfunkel, Z., and Nur, A., 1984, Block rotation by strike-slip faulting: Structural and paleomagnetic evidence: Journal of Geophysical Research, v. 89, p. 6256–6270.

Ron, H., Aydin, A., and Nur, A., 1986, Strike-slip faulting and block rotation in the Lake Mead fault system: Geology, v. 14, p. 1020–1023.

Speed, R.C., 1976, Geologic map of the Humboldt lopolith: Geological Society of America Map and Chart Series MC-14, scale 1:81,050.

Wernicke, B., 1981, Low-angle normal faults in the Basin and Range province—Nappe tectonics in an extending orogen: Nature, v. 291, p. 645–648.

ACKNOWLEDGMENTS

Supported by the Donors of the Petroleum Research Fund, administered by the American Chemical Society (16719-AC2 to Geissman), and a Geological Society of America research grant and an ARCO Fellowship to Hudson. Most of the paleomagnetic analyses were done at the U.S. Geological Survey (Denver) laboratory. We thank R. Reynolds, H. Ron, and J. Rosenbaum for fruitful discussions; R. E. Anderson, R. Reynolds, and N. J. Silberling for clarifying early drafts; R. F. Butler and an additional *Geology* reviewer for critical review of the manuscript; and M. Boone for manuscript preparation.

Manuscript received November 21, 1986
Revised manuscript received March 6, 1987
Manuscript accepted March 17, 1987